Akoizo Original Practice - Form 1

Division B | Hovercraft | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) TT is 10.0 s and the run takes 8.0 s. What is the Time Score?

A. 32

B. 8

C. 40

D. 20

Answer: _____

2. (2 points) Five runs at one setting give 11.8, 12.4, 11.6, 12.5, and 11.9 s. What should the team improve first?

A. The Target Time

B. The dowel height

C. Repeatability, because the spread is almost 1 s

D. The calibration curve slope

Answer: _____

3. (2 points) Four 1.2 V cells in series and two such strings in parallel give what voltage?

A. 2.4 V

B. 1.2 V

C. 4.8 V

D. 9.6 V

Answer: _____

4. (2 points) Which contact with the track is NOT allowed?

Unit: Levitation: the air cushion and the skirt.
Worked-case data: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?
Separate investigation observations:
Empty test: Vehicle glides freely across the floor with lift on.
With all nickels: Rear skirt drags and the craft stops quickly.
Nickels moved forward 6 cm: Glides freely again, sitting level.

A. The inside edge of the rails

B. Gentle skirt contact with the floor

C. The top of the side rails

D. The track surface

Answer: _____

5. (2 points) When does the team’s competition time end?

Unit: Impound, inspection, and competition day.
Worked-case data: A team’s vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

A. After two complete or two incomplete runs, or 8 minutes, whichever comes first

B. After one complete run

C. After 10 minutes

D. When the team decides

Answer: _____

6. (4 points) Calculate MS for a complete run carrying 2 rolls and 8 loose nickels.

Answer: _____
_____
_____

7. (4 points) Why does rail scraping hurt Time Score more than it hurts Distance Score?

Answer: _____
_____
_____

8. (4 points) Explain why an early finish is not better than a late one.

Unit: Scoring a run: Distance, Time, and Mass.
Worked-case data: TT is 12.0 s. A vehicle carries 3 rolls and 10 loose nickels and finishes in 13.2 s with nothing falling off. Find the run score.
Separate investigation observations:
Run plan A: No nickels, typical error 0.3 s at TT = 12 s, always completes.
Run plan B: All nickels, typical error 0.6 s, completes 9 times in 10.
Run plan C: All nickels with ramp, error 0.2 s.

Answer: _____
_____
_____

9. (4 points) How should a team use its first competition run to improve the second?

Answer: _____
_____
_____

10. (4 points) Why must a resistor used for speed control be rated for enough power?

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: A team's vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?

Answer: _____
_____
_____

13. (2 points) What was the main cost of the open tray?

Supplied synthetic observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

A. The construction penalty on the whole run

B. The 0.3 Mass Score points from three nickels

C. Nothing, because the run was complete

Answer: _____

14. (2 points) Define “Thrust” in the context of Hovercraft.

Answer: _____
_____
_____

15. (2 points) Define “Distance Score (DS)” in the context of Hovercraft.

Answer: _____
_____
_____

16. (2 points) Define “Interpolation” in the context of Hovercraft.

Answer: _____
_____
_____

17. (2 points) Define “Potentiometer” in the context of Hovercraft.

Answer: _____
_____
_____

18. (2 points) Define “Hover gap” in the context of Hovercraft.

Answer: _____
_____
_____